The evolution of center-embedding: Emergence, ephemerality, and processing constraints

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Drawing on older proposals, along with experimental and observational evidence, we test a hypothesis regarding the exaptation of linguistic recursion (in the form of center-embedding) from nonlinguistic activity, specifically, ritual behavior, as preserved in contemporary art forms. Center-embedding is comparatively reduced in language due to processing constraints imposed by working memory and discourse reference. In support of this theory, we look at patterned emoji usage on social media: center-embedded patterns are common, but become simplified or replaced with less complex repetitive patterns in transmission. Repetitious palindromes (e.g. ABABA) are also preferred over non-repetitious ones (e.g. ABCBA). In a pattern recognition experiment, we find that the recognizability of non-repetitious palindromes decreases with length, with a drop in accuracy around length 7. These results suggest that constraints on visual processing of nonlinguistic center-embedded sequences mirror those of verbal working memory for center-embedded sentences, with a similar cap on embedding depths.

1. Introduction: The exaptation and reduction of center-embedding

Hauser et al. (2002) suggested that syntactic recursion could have been exapted from nonlinguistic cognitive abilities (navigation, numerosity, social hierarchies) that are not unique to humans. In this study we pursue the hypothesis proposed by Staal (1979, 1980) that the origin of linguistic recursion is to be found in ritual practice; both he and Witzel (1992) emphasize its hierarchical center-embedded structure. Staal points to analogous structure in animal ritual as evidence for his hypothesis, and all animal studies of recursion to date test animals' ability to recognize or produce center embedding (Gentner et al. 2006, Abe & Watanabe 2011, Jiang et al. 2018, Ferrigno et al. 2020, Liao et al. 2022, Lameira et al. 2024). Center-embedded or palindromic structure is also pervasive in human culture, including music (Hofstadter, 1979; Rohrmeier et al., 2015) and narratives (Van Otterlo, 1944; Forte & Smith, 2014)—art forms that are historically and culturally linked to ritual (Minkowski, 1989; Merker, 2009). Likewise, mirrored symmetry is an important principle in many visual art forms, including ancient Indian architecture with its connections to Vedic ritual (Chakrabarti, 2013; Meister, 1983), in parallel to the symmetries found in temporal art forms, e.g. music and narrative (Porter, 1971). As we will discuss

further below, such patterns are also prevalent in online emoji use—a visual paralinguistic form of communication.

The ubiquity of center-embedded patterns in human culture, and their precedents in nonhuman cognition, raise the question of why center-embedding is relatively reduced in (especially spontaneous spoken/signed) language. Center-embedding is forced to the periphery of human language by processing pressures of time and limited working memory capacity (Chomsky & Miller, 1963; Miller & Isard, 1964; Gibson, 1998): sentences with more than two center-embedded clauses are generally difficult to comprehend and do not occur in real discourse (Karlsson, 2007). The fact that center-embedding is more prevalent in written than in spoken language (Karlsson 2009, 2020) provides further evidence for the role of such constraints in the limited nature of linguistic center-embedding.

On the other hand, as noted by Staal (1980), human ritual and performance art unfold on a time scale more conducive to center embedding, as they depend solely on collective long-term memory, often recorded and stored in external media. Similarly, center-embedded or palindromic sequences in physical media (architecture, sculpture, pottery, painting) are fixed, with no concomitant constraints on time (aside from physical deterioration) or memory. Emoji in social media present an interesting borderline case of a cultural artifact that, while visual in nature, is paralinguistic, i.e. meaning-bearing in a way that supplements ephemerally written language (Grosz et al., 2023). These digital communications unfold on a timescale of minutes or less—while they may persist indefinitely online, in practice they are scrolled past rapidly. As we show in Section 2, palindromic sequences are also prevalent in emoji strings on social media, but these tend to be short and composed of alternating characters, and may be simplified or restructured in transmission. In the experiment described in Section 3, we speeded up the appearance of emoji strings to approach timescales closer to those of linguistic processing, in order to determine if, under time pressure, constraints on levels of embedding similar to those found in real-time linguistic processing also emerge in the visual processing of emoji strings.

2. Embedded emoji patterns: emergence and disappearance

Symmetrical center-embedded strings of emoji are widely used on social media, where they may be used to bracket words, adding emphasis or tone (McCulloch, 2020:127), or to supplement or illustrate the text (McDonald, 2024). A few examples we have observed online are shown in (1):

(1)
(a) 🤐❌👮❌🤐
(b) 😛🌪️⚡🌪️😛
(c) 🦁🐯🐻 MYSTERY CRITTER 🐻🐯🦁
(d) 💖🕊️☮️🌾🌻🇺🇦☀️🙏🏻🙏🏻 SLAVA UKRAINI HEROYAM SLAVA🙏🏻🙏🏻☀️🇺🇦🌻🌾☮️🕊️💖

We set out to quantify the occurrence of such patterns by searching a publicly available Twitter corpus (Ma, 2017) containing ~5 million tweets from 2016-2017, and extracted those containing multiple emojis. These strings were in turn classified into the patterns shown in Figure 1.

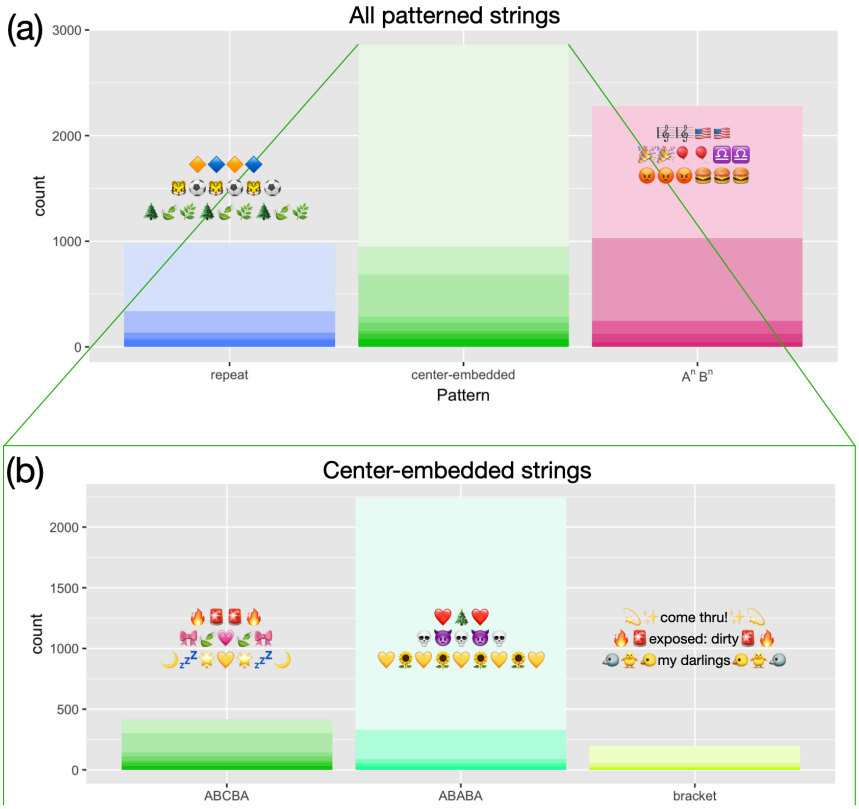

Figure 1: Frequency of emoji string patterns in Twitter corpus (Ma 2017): all patterns (a), and center-embedded strings (b), with examples of each. Darker color indicates greater length.

There was a clear preference for mirror symmetry: 2701 of 22,560 multi-emoji strings (12%) were palindromes (1a&b) or brackets around text (1c&d); on the other hand, 2035 (9%) showed an A^nB^n pattern (Figure 1a, right column), and

981 (4%) showed a repeating pattern (Figure 1a, left column). 83% of the palindromic strings, however, consisted of alternating characters (e.g. ABABA, Figure 1b, middle column) and 65% of the non-alternating palindromic strings (e.g. ABCBA, ABBA, Figure 1b, left column) were of length 5 or shorter, suggesting that this preference for palindromes is counterbalanced by a tendency for brevity and simplicity.

In addition, informal observations of group text and Zoom chat threads in which we were included showed the emergence of chains of palindromic emoji strings accompanying text. These chains demonstrated the evolution of such strings through informal transmission: strings were picked up in modified form by other group members, often preserving the same or similar emojis in a simpler pattern (Figure 2). As seen in the chain in the right half of Figure 2, the ABABABA palindrome gives rise to repeat strings (ABABABAB and ABAB).

Figure 2: Evolution of emoji strings in text conversations (text removed). Green arrows indicate palindromic strings.

These observations are consistent with experimental work showing that center-embedded sequences are particularly difficult to learn, even compared to the more computationally complex pattern of cross-serial dependencies (Moreton et al., 2021; Ferrigno et al., 2025), and that these patterns tend to vanish in iterated transmission (Davis, 2023b). In meaningful artificial languages, center-embedding is also more difficult to learn than branching patterns, and does not emerge in iterated transmission, even though cross-serial patterns occasionally do (Davis & Smith, 2023).

The preceding quantitative and qualitative observations of emoji usage, along with sequence learning, suggest that while palindromes seem to be

preferred over other patterns, they are unstable and easily replaced with simpler patterns in real time, reflecting constraints that also operate in spontaneous language. In addition, there is a preference for shorter and/or repeat palindromes, which are likely simpler to generate and recognize. Keeping track of center-embedded patterns in a string that is viewed briefly, such as in a text thread, and is too long to easily process visually, may contribute to the elimination or restructuring of palindromes in transmission. We further investigated these factors in a pattern-recognition experiment.

3. Palindrome pattern recognition experiment

3.1. Introduction

In working with emoji strings like those in (1), we noticed that with very long strings (such as 1d), it is no longer possible to tell at a glance if a string is actually palindromic. At that point, one instead has to adopt a visual strategy of serial search for matching icon pairs. This reminded us of the offline "pencil and paper" analysis of matching subjects and predicates that becomes necessary when linguistic center-embedding gets too complicated. We therefore sought to determine the point at which an emoji string can no longer be processed as a whole Gestalt, in order to test if purely visual processing is subject to constraints on levels of embedding similar to those operative in reading linguistic material. We deliberately set the timing parameters of our experiment to prevent saccades. In pilot work, we determined that a presentation duration of 700 ms gave participants enough time to process an emoji string as a whole without allowing them to adopt a serial search strategy for matching icon pairs.

3.2. Methods

We conducted this experiment online with English-speaking participants (n = 60) recruited through Prolific. Emoji strings were generated from a set of seven emojis chosen to be visually distinctive in shape and color (Figure 3). The stimuli included palindromic strings of four lengths: 3, 5, 7, or 9 emojis; for strings greater than length 3, both "alternating" (e.g. ABABA, ABABABA) and non-alternating palindromes (e.g. ABCBA, ABCDCBA) were used, as well as repeat strings (ABABAB). There were also "false palindrome" strings created by modifying the non-alternating palindromic strings as follows: deleting one emoji (ABCD[_]BA), exchanging two adjacent emoji (ABCD**BC**A), or replacing one emoji with another that did not occur in the string for lengths 3 and 5 (ABA →AB**C**, ABCBA→ABC**D**A), or that occurred earlier in the string for longer palindromes (ABCD**A**BA). These conditions were designed to avoid adjacent repetition of emoji (e.g. ABCAA), which would make non-palindromes

too easy to recognize based on local criteria. In each trial, participants viewed a fixation cross for 2000 ms, then a string of emojis for 700 ms, and were asked to identify whether it was a palindrome. Each participant performed 80 trials, with four strings of each type. Trials were shown in random order.

3.3. Results

The results were analyzed using a mixed-effects model with string length and string type—palindrome vs. non-palindrome and alternating (ABABA) vs. non-alternating palindrome (ABCBA)—as fixed effects, and participant as a random effect.

As shown in Figure 3, there is a striking difference between alternating and non-alternating palindromic strings: alternating strings were easier to recognize correctly as palindromes and were thus at ceiling regardless of length, while accuracy in recognizing non-alternating palindromes was lower and declined with length ($B = -0.04$, $SE = 0.001$, $p < .0001$). There was also a significant effect of length for both ABCBA palindromes ($B = -0.04$, $SE = 0.01$, $p < .0001$) and false palindromes ($B = -0.09$, $SE = 0.005$, $p < .0001$), i.e. longer strings were less likely to be identified correctly. Finally, participants recognized palindromes more accurately than non-palindromes across all conditions ($B = 0.35$, $SE = 0.1$, $p < .001$). Although false positives were common, participants were generally capable of accurately identifying palindromes and rejecting false palindromes and repeat strings ($d' = 1.226$).

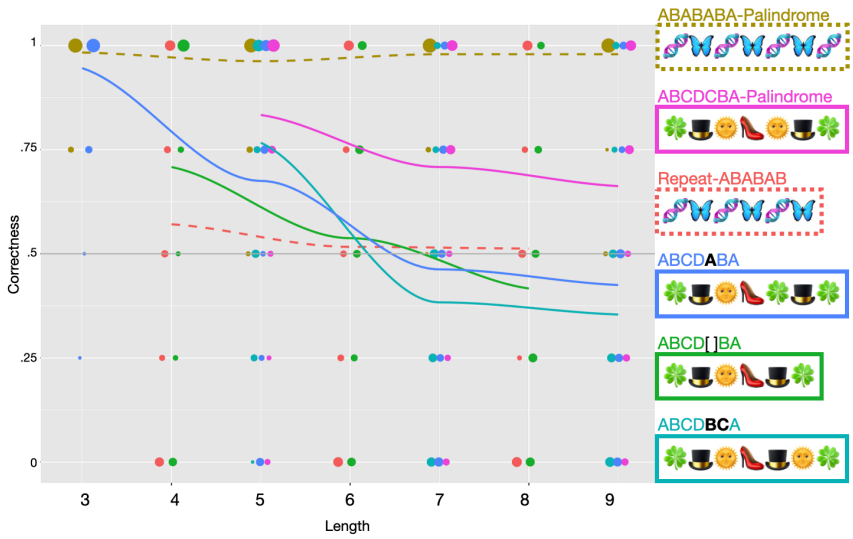

Figure 3. Results of pattern recognition experiment by string length and type, with examples.

3.4. Discussion

In some respects, our results seem unremarkable: there is a clear drop-off in participants' ability to identify false palindromes between strings of length 5 (two levels of embedding, 70-75% accuracy) and strings of length 7 or 9 (three or more levels of embedding, below chance). As we expected, speeded presentations of five-emoji palindromes can still be processed holistically and categorized accurately, while longer sequences require serial search for matched pairs in appropriate serial positions, which our experimental paradigm disallowed. This drop-off did not occur with alternating ABABABA palindromes, which were consistently near ceiling. These patterns may be recognizable as a Gestalt—the alternating pattern is clearly visible and its status as a palindrome can be determined from the fact that the first and last emojis match. Performance on repeat strings (ABABAB) was near chance, possibly because these strings were easily confused with the similar-looking alternating palindromes.

Thus, on the one hand, this result seems consistent with proposed constraints on permissible levels of center embedding in language processing (Karlsson 2007) and on working memory in general (Miller 1956). On the other hand, this was not a language processing task, and there was no visual working memory component to the task, either: participants were either able to take in the entire string at an exposure of 700 ms or not. This suggests that human limitations on the processing of center-embedded sequences apply to visual pattern recognition as well, and are therefore more domain-general in nature.

Our larger point, however, is that speeding up the presentation of emoji strings to approach the speed of language processing (though still above it by several hundred milliseconds) renders center-embedded sequences as difficult to process visually as they are linguistically. For example, palindromic strings from which a requisite emoji was deleted (e.g. ABCDBA) predominantly elicited false alarms. This is akin to the “missing VP” phenomenon in language processing, in which the predicate to an intermediate center-embedded subject goes entirely missing without participants noticing (Gibson & Thomas, 1999). This is precisely why we chose to target the processing of emoji palindromic strings in the first place: they straddle the cusp between visual and linguistic processing, and as such open up an avenue for further testing of Staal's (1979, 1980) hypothesis: recursion in the form of center embedding was exapted for linguistic purposes from ritual practice and its cultural and artistic offshoots.

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